Faces or Facts?

Experimental Evidence on Time Pressure, Information Costs, and Gender Bias in Electoral Choice

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Abstract

Voters often see a candidate's face before learning about the candidate's political positions. Building on evidence that rapid visual impressions predict electoral outcomes, we examine how time pressure and information costs shape political learning and whether these constraints produce distinct patterns of scrutiny for women candidates. We conduct a preregistered three-arm online visual conjoint experiment using photographs and political information drawn from Colombian electoral data. Respondents are randomly assigned to make these choices under a 20-second deadline, an effort cost for accessing political information, or an unconstrained baseline. Candidate pairings, task order, and left-right placement are randomized within respondents. The experiment measures recorded candidate selection, candidate-specific information acquisition, and decision time. It identifies intention-to-treat effects of the decision environment and causal effects of assigning complete candidate profiles. Because the attributes of real candidates are not independently randomized, analyses of gender, ideology, and appearance characterize heterogeneity across observed profile bundles rather than attribute-level average marginal component effects.

1 Faces arrive before facts

Most political choices begin with a sequence so ordinary that it is easy to overlook: voters see a face before they read a platform. A visual impression arrives almost automatically, whereas learning a candidate's ideology and priorities requires time and effort. This ordering matters because judgments made from faces can form after exposures as brief as 100 milliseconds and predict real election outcomes (Todorov et al. 2005). The familiar contrast between "faces" and "facts" is therefore not merely rhetorical. It describes two sources of political judgment that become available at different speeds and costs.

Time pressure matters because it changes how decision makers acquire and combine information before choosing. When time is scarce, people accelerate processing, filter cues, or substitute a simpler noncompensatory rule for effortful integration (Payne, Bettman, and Luce 1996; Rieskamp and Hoffrage 2008). Political learning follows the same basic logic. Candidate evaluations can be updated on-line as information is encountered, even when the facts supporting the resulting impression are poorly remembered (Lodge, McGraw, and Stroh 1989). A deadline can therefore alter which evidence enters the running evaluation and how extensively competing considerations are compared before a vote.

Antonakis and Dalgas (2009) make this puzzle unusually concrete. After an imaginary voyage from Troy to Ithaca, Swiss children aged 5–13 were shown pairs of photographs from French parliamentary runoffs and asked whom they would choose as captain. Their choices predicted the actual winner with a probability of 0.71; adults' competence ratings predicted the winner with a probability of 0.72, and the child and adult response patterns were statistically indistinguishable. The participants did not need

campaign knowledge, policy positions, or electoral experience to reproduce the electorate's ranking. As Antonakis and Dalgas put it, "voters are anchored in first impressions and do not appropriately correct initial inferences" (2009, 1183). Ballew and Todorov (2007) identify the speed of this process more directly: competence judgments after 100 or 250 milliseconds were as predictive as judgments made with unlimited exposure, and a two-second response deadline preserved predictive accuracy while instructions to deliberate lengthened response time and reduced it.

That finding is compelling, but it leaves the central decision process unobserved. We know that faces can predict choices; we know much less about when voters stop at the face, when they seek political information, and whether constraints make one source dominate the other. Candidate photographs operate as cues in low-information elections (Banducci et al. 2008), appearance predicts success across political contexts (Rosar et al. 2008; Berggren et al. 2010; Lawson et al. 2010), and media exposure can strengthen appearance-based voting among less informed citizens (Lenz and Lawson 2011). Faces also convey perceived competence, ideology, party, status, and threat rather than attractiveness alone (Mattes et al. 2010; Herrmann and Shikano 2016; Olivola et al. 2018).

1.1 Gendered scrutiny and the cost of learning

Candidate gender can shape not only the final vote but the path by which voters learn. Ditonto, Hamilton, and Redlawsk (2014) show that voters seek more competence-related and "compassion issue" information about women than men. Andersen and Ditonto (2020) further show that the amount of search triggered by a woman candidate depends on partisan alignment and is strongest early in a campaign. This is consequential because women are often required to provide stronger evidence of qualification: across survey experiments, voters applied more stringent qualification standards to women candidates (Bauer 2020). A photograph makes candidate gender immediately available before policy text is opened, exactly the sequence in which an early cue can organize subsequent search.

The mechanism need not involve conscious hostility. Campaign communication can activate gender stereotypes that otherwise remain dormant (Bauer 2015), and rapid appearance-based inferences predict different outcomes by candidate gender: competence is most predictive in men-only races, whereas attractiveness predicts women-only races (Ditonto and Mattes 2018). Bahamonde and Sarpila (2024) likewise find a compounded penalty for working-class-looking women and weaker electoral returns to attractiveness for women than men. Recent work cautions against a universal penalty: eye-tracking evidence finds similar attention to stereotype-consistent and inconsistent information across candidate gender, suggesting that bias can arise from the standards applied to evidence rather than from attention alone (Jenke 2025), while think-aloud evidence shows both rapid gender heuristics and later gendered rationalizations (Rohrbach and Schönhagen 2025). The empirical question is therefore not simply whether women lose. It is whether profiles of women candidates invite different patterns of information acquisition, whether constraints suppress potentially corrective learning, and whether the same evidence is translated into choice differently for women and men.

Taken together, these findings motivate a design that follows the decision process from initial exposure to information acquisition and, ultimately, choice. Every respondent encounters the same paired-candidate task, while random assignment determines whether the choice occurs under a 20-second deadline, whether political information is costly to access, or whether neither constraint applies. Candidate-specific information requests reveal whether these constraints alter both the amount and the gendered distribution of political learning. The study thus locates inequality within a sequence—from first impression, to information acquisition, to choice—rather than only in the final vote. Methodologically, it combines randomized time and effort constraints with behavioral measures of information acquisition and

repeated choices between profiles constructed from Colombian electoral data.[1] Standard conjoint average marginal component effects require independently randomized attributes (Hainmueller, Hopkins, and Yamamoto 2014). Here, photographs, gender, ideology, electoral history, and facial characteristics remain bundled within real candidate profiles. The primary causal claims therefore concern the randomized decision environment and the assignment of complete profiles. Analyses of individual profile characteristics assess heterogeneity in those effects.

The first contribution is to recast appearance-based voting as a problem of information acquisition and evidence integration rather than simply an association between facial impressions and electoral choice. The same observed vote can emerge from several distinct processes: a voter may never consult political information, may inspect only one candidate's positions, may examine both candidates yet retain the initial visual ranking, or may use programmatic evidence to revise that ranking. These paths carry different theoretical and practical implications, but studies that observe only trait judgments or final votes cannot distinguish among them. By measuring candidate-specific information requests, decision time, and choice across repeated tasks, the experiment separates the decision to acquire political information from the electoral decision made in its presence or absence. The comparison among a deadline, an access cost, and an unconstrained baseline then reveals whether temporal and effort constraints reduce learning altogether, make learning asymmetric across candidates, or alter choice even when average acquisition changes little. The design therefore moves beyond showing that faces matter: it identifies the decision environments under which voters acquire evidence capable of supplementing an initial visual impression and estimates whether those environments alter candidate choice.

The second contribution is to connect this process account to gender inequality in candidate evaluation. More extensive information search about women candidates may reflect genuine openness to learning, but it may also indicate that women must supply additional evidence before voters regard them as qualified; conversely, limited search may signal either confidence or exclusion from serious consideration (Ditonto, Hamilton, and Redlawsk 2014; Bauer 2020). A gender difference in information acquisition therefore has no invariant meaning apart from the conditions under which it arises and the choices that follow it. Randomizing the decision environment supplies the relevant leverage. If the difference between women and men candidates expands, contracts, or reverses under time pressure or costly access, the experiment identifies how those constraints redistribute scrutiny across fixed profile groups, without treating candidate gender as independently randomized. Linking candidate-specific acquisition to candidate selection further documents whether unequal scrutiny is accompanied by unequal electoral returns. This contribution is consequential beyond the experimental interface: campaigns, ballot environments, and digital information platforms routinely vary in the time and effort they require from voters. Establishing whether such frictions place unequal informational burdens on women candidates connects a micro-level process of political learning to broader inequalities in evaluation and representation.

2 Research design

2.1 Sample and eligibility

The planned sample consists of 462 adults aged 18–65, balanced across men and women and allocated approximately equally across the three experimental conditions (154 per condition).[2] To minimize candidate recognition and prior knowledge of Colombian politics, respondents must not be Colombian nationals, reside in Colombia, or have previously lived in Colombia. Country of residence and nationality

[1] The study is registered in the AEA RCT Registry, AEARCTR-0017574, version 1.3, https://doi.org/10.1257/rct.17574-1.3. IRB Approval Number 086-2025. This paper describes the protocol following pretesting, including the revised 20-second deadline and fifteen analytical tasks. The registration's sample-size fields specify 444 participants and eight tasks, whereas its abstract specifies 462 participants. The current protocol specifies 462 participants and randomizes complete profiles rather than individual attributes.

[2] Participants are recruited through the survey company Borde Data (https://www.bordedata.cl). An initial recruitment wave comprises 30 eligible completed interviews, which are retained for analysis and included in the total sample of 462. The remaining recruitment target is therefore 432 additional completions.

are selected from country lists, followed by a question on previous residence in Colombia. Respondents who report a Colombian connection or decline to answer the residence-history question leave the study before the choice tasks. Eligibility screening and the baseline questionnaire precede exposure to the assigned-condition instructions. The questionnaire measures age, gender identity, occupation, education, electoral participation, political interest, frequency of political discussion, and ideological self-placement. Self-placement uses a 1–10 scale, anchored at left and right, with separate options for uncertainty, nonresponse, and identification with neither side.

2.2 Stimuli and repeated choice

After the pretreatment questionnaire, respondents complete repeated paired choices drawn from 71 Colombian candidate profiles for which a photograph and political information are available. For each respondent, 40 profiles are sampled without replacement and randomly ordered. Consecutive profiles form 20 pairs, randomizing both pair composition and task order. The first five pairs are used for practice, and the remaining fifteen constitute the analytical sample. Within each task, the two candidates are assigned to left and right by an independent random permutation of the pair. No candidate therefore appears more than once for the same respondent, although candidates can appear for different respondents. Both photographs become visible together after loading. Descriptions are initially hidden and identify each candidate's ideological orientation as left, center, or right, followed by the candidate's thematic priorities. Respondents may reveal either description and revise their selection until they submit the task or the deadline expires. Information can therefore be accessed before or after an initial selection. The five practice rounds reproduce the assigned condition, including its deadline or arithmetic requirement, and are excluded from the analytical sample.

The fifteen-task analytical sequence is supported by evidence on the response quality of repeated conjoint choices. Across six online experiments involving 6,534 respondents recruited through Amazon's Mechanical Turk and Survey Sampling International, Bansak et al. (2018) examined thirty paired candidate choices per respondent. They found only limited increases in survey satisficing. The estimated effects of candidate education and partisanship declined slightly between the first and second tasks but remained broadly stable across the remaining sequence, as did the explanatory power of these attributes. Fifteen analytical choices, and twenty choices when practice is included, therefore fall within a range over which repeated candidate evaluations have retained substantial informational value. Because the appropriate task burden also depends on the full instrument, we will examine response times, information requests, nonresponse, and choice patterns across task positions to assess fatigue under the additional demands of photographic evaluation, experimental constraints, and follow-up questions.

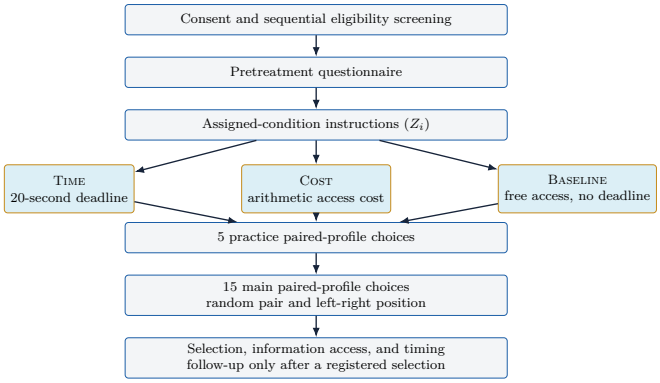
After each analytical task with a registered selection, respondents are asked whether they would vote for the candidate they have just chosen in a real election. Those answering affirmatively identify the principal considerations underlying their choice. The eligible response options are randomized within each task, with "Otra razón" always placed last. The questions themselves retain a fixed order. The factor list includes appearance-related considerations, while ideology and thematic priorities appear only if at least one candidate description was revealed. Perceived time pressure is measured with the same seven-point item in Time and Baseline, regardless of the answer to the real-election support question. Perceived effort is measured on a seven-point scale in Cost only when the respondent revealed information. Tasks that end without a selection skip every follow-up question.

2.3 Experimental conditions

Let $Z_i \in \{T, C, B\}$ denote assignment to one of three decision environments. In Time, each task includes a visible 20-second countdown, while the candidate descriptions can be opened without an arithmetic requirement. In Cost, there is no deadline, but revealing each candidate's description requires correctly solving a simple addition problem. In Baseline, neither a deadline nor an arithmetic barrier applies. Allocation occurs with equal probability at the respondent level when study sessions are initialized and remains concealed through screening and the baseline questionnaire. The candidates, number of tasks,



A. Example choice task. Respondents compare two Colombian candidates and may open political information before submitting a selection. Information for the candidate on the left is shown here.



B. Assignment and measurement. Assignment is concealed until the condition-specific instructions and remains fixed during practice and analytical tasks. Each respondent receives an independent 40-profile sequence sampled without replacement. Pair order and left-right placement are randomized.

Figure 1: Experimental sequence and example choice task.

and choice prompt are otherwise held constant. This structure distinguishes two constraints on political learning. Time limits the time available to acquire and integrate information, whereas Cost increases the effort required to obtain it. In every condition, decision timing begins only when both photographs have loaded and decoded successfully and are revealed together. Loading time is recorded separately, and an image-loading failure prevents the task from starting until the images are available. The same 20-second deadline applies during practice and analytical tasks in Time. If the deadline expires, the current selection is submitted automatically. When no candidate has been selected, the choice remains missing.

2.4 Temporal scarcity and deadline calibration

The 20-second deadline is designed to restrict opportunities for political learning while retaining the possibility of consulting substantive information. Ballew and Todorov (2007) show that competence judgments based on candidate photographs predict electoral outcomes after exposures of 100–250 milliseconds and under a two-second response deadline. Their task involves a judgment from faces alone, whereas the present experiment requires voters to decide whether to obtain political information and how to incorporate it into an electoral choice. This distinction matters because temporal scarcity can change not only response speed but also the evidence that enters a decision. Time-constrained choice encourages selective search and simpler rules for integrating information (Payne, Bettman, and Luce 1996; Rieskamp and Hoffrage 2008), and opportunity costs of decision time can alter voting behavior (Alós-Ferrer and Garagnani 2022). Interacting with the interface also requires motor actions, cognitive preparation, and system response, whose durations depend on the task and interface rather than a universal time per click (Card, Moran, and Newell 1980; Sauro 2009). The treatment therefore imposes scarcity on the complete sequence of information acquisition, comparison, and choice rather than attempting to isolate an instantaneous facial judgment.

Calibration of this decision window accounts for the amount of information available and the time required to use it. Each candidate description contains 10–17 Spanish words, with a median of 14, so consulting two median-length descriptions entails reading 28 words in total. Reading rates vary across readers and materials (Brysbaert 2019). Among two groups of Chilean law students reading different legal texts ($N = 200$), Caballero Germain et al. (2021) report mean rates of approximately 178 and 228 words per minute. Applying these contextual benchmarks to 28 words implies approximately 7.4–9.4 seconds for reading alone, leaving part of a twenty-second window for accessing the descriptions, inspecting the photographs, comparing candidates, and registering a selection. These calculations are illustrative rather

than estimates of comprehension or guarantees for every respondent. Pretest feedback indicated that the initial ten-second deadline left some testers insufficient time to consult the descriptions, motivating its extension to twenty seconds. The resulting limit is thus a task-specific, pretest-informed restriction that is intended to preserve meaningful information search while constraining its duration. Responses from development pretesting conducted before fieldwork and the five within-respondent practice tasks are excluded from the analytical sample. The initial recruitment wave remains part of the fieldwork sample. The effectiveness of the deadline will be assessed through information acquisition, nonselection at the deadline, and comparisons with the untimed baseline in task duration and reported hurry. Because the deadline mechanically caps task duration, faster responses alone will not be treated as evidence that respondents experienced pressure. Reported hurry is observed only after a registered selection, so this comparison will be interpreted alongside differential nonselection rather than as an unconditional measure of the manipulation's effect.

2.5 Outcomes and measurement

The primary outcomes are recorded candidate selection, whether any description is revealed, the number of descriptions revealed, time to first selection capped at a common twenty-second window, and failure to register a selection within that window. Description revelation is the observed measure of information acquisition and does not establish that the text was read or understood. These candidate-specific records permit tests of gendered scrutiny. Total time from photograph display to task submission, changes in selection, and information-button presses are recorded separately. Information-acquisition measures cover the entire task, including openings after an initial selection. Terminal nonselection at the deadline is retained as a separate outcome. Post-choice measures capture whether respondents would support the selected candidate in a real election and which considerations they report using. The candidate profiles include gender, age, party affiliation, geographic constituency, electoral performance, political information, and previously measured facial characteristics. Recording the alternatives and information available in each task allows us to distinguish treatment effects on information acquisition from treatment effects on recorded selection.

3 Potential outcomes and estimands

3.1 Notation and common averaging distribution

For eligible respondent i in analytical task t, let $Z_i \in \{T, C, B\}$ denote the assigned decision environment and let $P_{it} = (C_{it}^L, C_{it}^R)$ denote the ordered candidate pair. For side $s \in \{L, R\}$, define $M_{it}^s(z, p) \in \{0, 1\}$ as the potential indicator that the respondent reveals the political description of the candidate on side s at any point before the task ends under environment z and pair p. This measures access to the text, without establishing that it was read or understood. Respondents may reveal information after an initial selection and revise that selection before submission. Aggregate information acquisition is defined as

$$K_{it}(z,p) = M_{it}^L(z,p) + M_{it}^R(z,p), \quad A_{it}(z,p) = \mathbf{1}\{K_{it}(z,p) \geq 1\}, \quad X_{it}(z,p) = \mathbf{1}\{K_{it}(z,p) = 2\}.$$

Thus, K is the number of descriptions revealed, A indicates access to at least one description, and X indicates access to both descriptions.

Let $T_{it}^*(z,p)$ denote seconds from the simultaneous display of both fully loaded photographs to the first candidate selection, with $T^* = \infty$ if no selection occurs. To place all conditions on a common observation window, define capped selection latency $D_{it}^{20}(z,p) = \min\{T_{it}^*(z,p), 20\}$ and slow or absent selection $S_{it}^{20}(z,p) = \mathbf{1}\{T_{it}^*(z,p) > 20\}$. If no selection occurs within twenty seconds, $D^{20} = 20$ and $S^{20} = 1$. The full task duration, measured from photograph display to submission, is recorded separately and can include information acquisition and revisions after the first selection. Loading time precedes both measures. Let $Q_{it}(z,p)$ indicate that a task terminates at the deadline without any candidate selection. This outcome is possible under TIME; the other conditions require a selection before submission. Reaching the deadline after selecting a candidate preserves the current selection and gives $Q = 0$. Thus,

S^{20} measures failure to select within the common window, while Q identifies termination without a candidate response.

Candidate choice is defined without privileging the randomized left position. Let $V_{it}(z,p)$ denote the candidate selected when the task ends; $V = \varnothing$ denotes a missing candidate response when $Q = 1$. For each displayed side, define the registered-selection event

$$Y_{it}^{s}(z,p) = \mathbf{1}\{V_{it}(z,p) = C_{it}^{s}\}.$$

When a candidate response is recorded, $Y_{it}^{L} + Y_{it}^{R} = 1$; when $Q_{it} = 1$, both derived event indicators equal zero, while the candidate response itself remains missing. This definition neither imputes a vote nor assigns an unobserved preference to a respondent who made no selection. It makes Y^s the intention-to-treat probability that a displayed profile receives a registered selection under the assigned environment. Effects on this outcome can reflect both changes in which candidate is selected and changes in whether any selection is completed.

Unless otherwise stated, expectations weight eligible respondents and the fifteen analytical tasks equally and average over the randomized candidate draw, pairing, task order, and left-right assignment. Practice tasks are excluded. Comparisons between $z \in \{T, C\}$ and BASELINE are respondent-level intention-to-treat effects.

3.2 Effects of the decision environment

For each task-level outcome $R \in \{A, K, X, D^{20}, S^{20}\}$, define

$$\tau_{zB}^{R} = \mathbb{E}[R_{it}(z, P_{it}) - R_{it}(B, P_{it})], \qquad z \in \{T, C\}.$$

The terminal-timeout effect is defined separately as $\tau_{TB}^{Q} = \mathbb{E}[Q_{it}(T, P_{it}) - Q_{it}(B, P_{it})]$ because forced task termination exists only in TIME. These estimands capture how temporal scarcity and information cost change information acquisition, decision speed, and response completion. Information acquisition is also a potential mediator of candidate choice. Because it occurs after treatment, models that condition choice on realized information acquisition do not identify a controlled or natural direct effect without additional assumptions; mediation analyses are therefore interpreted cautiously and do not carry a direct-effect interpretation.

Randomized pairing and position also identify side-neutral profile-selection probabilities. For complete candidate profile c, define

$$\theta_c(z) = \mathbb{E}[Y_{it}^{s}(z, P_{it}) \mid C_{it}^{s} = c], \qquad \Delta_c(z) = \theta_c(z) - \theta_c(B).$$

The expectation averages over respondents, tasks, randomized rivals, and side. $\Delta_c(z)$ is the causal effect of the decision environment on the probability that profile c is selected when displayed. Contrasts $\theta_c(z) - \theta_{c'}(z)$ compare randomly assigned complete profiles under the same environment. They do not isolate gender, ideology, party, or facial characteristics because those attributes remain bundled within each observed candidate.

3.3 Gendered information acquisition and choice

Let $G(c) = 1$ denote a profile coded as a woman. For $R \in \{M, Y\}$, define the candidate-level mean, gender-indexed contrast, and treatment-induced change as

$$\mu_{g}^{R}(z) = \mathbb{E}[R_{it}^{s}(z, P_{it}) \mid G(C_{it}^{s}) = g], \quad \Gamma_R(z) = \mu_1^R(z) - \mu_0^R(z), \quad \Psi_R(z) = \Gamma_R(z) - \Gamma_R(B).$$

$\Gamma_M(z)$ compares the probability of consulting political information about women and men candidates, while $\Gamma_Y(z)$ compares their probability of selection. The corresponding $\Psi_M(z)$ and $\Psi_Y(z)$ estimands identify how random assignment to a constrained decision environment changes each gender-indexed contrast. Candidate gender is not independently randomized, so $\Gamma_M(z)$ and $\Gamma_Y(z)$ compare complete profiles grouped by gender and are not causal effects of gender. The causal interpretation of $\Psi_M(z)$ and $\Psi_Y(z)$ is limited to the effect of the randomized environment on those fixed profile-group differences.

3.4 Hypotheses and analytical priorities

The **primary process family** consists of τ_{zB}^{A}, τ_{zB}^{K}, $\tau_{zB}^{D^{20}}$, and $\tau_{zB}^{S^{20}}$ for $z \in \{T, C\}$. The **primary causal gender family** consists of $\Psi_M(T)$, $\Psi_M(C)$, $\Psi_Y(T)$, and $\Psi_Y(C)$. The baseline contrasts $\Gamma_M(B)$ and $\Gamma_Y(B)$ form a separate descriptive family because candidate gender is not independently assigned. Hypothesis tests are two-sided even where theory implies a direction.

We expect both constrained conditions to reduce any information acquisition and the number of descriptions consulted, $\tau_{zB}^{A} < 0$ and $\tau_{zB}^{K} < 0$ (H1). We also expect TIME to reduce capped selection latency, $\tau_{TB}^{D^{20}} < 0$, while effects on S^{20} and the timing effects of COST are tested without directional restrictions (H2). Under BASELINE, we expect respondents to consult information more often for profiles of women candidates, $\Gamma_M(B) > 0$, and both constraints to attenuate this difference, $\Psi_M(T) < 0$ and $\Psi_M(C) < 0$ (H3). Finally, we expect $\Psi_Y(T)$ and $\Psi_Y(C)$ to differ from zero, with their signs left open because prior work implies both potential penalties and advantages for women candidates (H4).

Additional estimands include acquisition of both descriptions X, terminal timeout Q, full task duration, profile-specific $\theta_c(z)$ and $\Delta_c(z)$, the real-election support item, self-reported time pressure and information cost, ideological congruence, associations between choice and other candidate characteristics, and moderation by respondent gender or political sophistication. **Additional analyses** examine mediation through information acquisition, models conditioned on realized information acquisition or completed choice, high-dimensional profile interactions, decision-factor responses, and subgroup variation. Because these analyses require further assumptions or extend beyond the central hypotheses, they will be interpreted separately from the primary tests.

3.5 Timeout and missing-data rules

All descriptions revealed before a task terminates remain observed and contribute to M^s, A, K, and X, including when no candidate is selected. A task with no selection within twenty seconds receives $D^{20} = 20$ and $S^{20} = 1$. If the TIME task terminates without a selection, $Q = 1$: the recorded candidate response remains missing, the derived registered-selection indicators satisfy $Y^L = Y^R = 0$, and every follow-up question is skipped. These undefined follow-up responses will not be imputed. If a selection has already occurred when the deadline expires, the current candidate response is retained, $Q = 0$, and the usual follow-up questions are shown. Analyses restricted to tasks with $Q = 0$ will be reported only as secondary sensitivity analyses; because completion can be affected by treatment, these comparisons do not by themselves identify the full intention-to-treat effect. Withdrawal or abandonment before task submission will remain missing rather than being recoded as a timeout; the number of analytical tasks completed and study attrition will be reported by condition. No other primary outcome will be imputed.

4 Estimation strategy

The analysis population includes all eligible randomized respondents who begin the experimental task sequence; no respondent will be excluded because of post-assignment behavior. Submitted analytical tasks contribute to the task-level outcomes, while unsubmitted tasks remain missing and contribute to the attrition analysis. Outcomes A, K, D^{20}, and S^{20} will be estimated using linear models with indicators for TIME and COST; the coefficients provide unadjusted assignment contrasts relative to BASELINE for observed tasks. Interpreting these estimates as intention-to-treat effects for the full analysis population requires assumptions about outcomes in missing tasks; randomization alone does not resolve informative attrition. Standard errors will be clustered by respondent. Covariate-adjusted specifications using pretreatment characteristics and task-order indicators will be reported as precision-adjusted robustness checks rather than replacing the unadjusted estimates.

Candidate-level analyses use two observations per submitted task, one for each displayed candidate, including tasks that terminate without a selection. Linear probability models for M^s and the derived registered-selection outcome Y^s include treatment indicators, candidate gender, treatment-by-gender interactions, randomized side, and task-order indicators. The interaction coefficients estimate $\Psi_M(z)$ and

$\Psi_Y(z)$, subject to the same qualification concerning missing tasks. Primary uncertainty for these models will use two-way clustering by respondent and candidate; respondent-clustered estimates and crossed random-intercept models will be reported as robustness checks. Profile-specific selection probabilities will be presented with multiplicity-adjusted uncertainty and will not be interpreted as attribute-level average marginal component effects.

Within the primary process and causal gender families, two-sided p-values will be adjusted separately by Holm's sequential procedure. The two baseline gender contrasts constitute a third, descriptive multiplicity family. Unadjusted estimates with 95% confidence intervals will also be reported. Balance checks assess pretreatment covariates and the randomized distribution of candidate profiles and positions across conditions. Manipulation checks examine arithmetic-problem completion, information acquisition, D^{20}, S^{20}, and Q, together with reported hurry in TIME relative to BASELINE among tasks with a registered selection. The latter comparison is subject to the completion-related selection described above. We will report screening, attrition, and task completion by condition; examine practice-to-analysis learning and task-order patterns; and distinguish candidate gender from respondent gender throughout. The analysis will follow the treatment definition, pretest-informed deadline, sample, estimands, hypotheses, exclusion rules, timeout coding, multiplicity adjustments, and estimation procedures described above.

5 Discussion

The experiment identifies whether randomized constraints alter political information acquisition, candidate choice, and the scrutiny applied to women candidates. Because gender, ideology, party, and facial characteristics remain bundled within observed candidates, these comparisons capture heterogeneity across naturally occurring profiles rather than isolated attribute effects. Observed information acquisition locates a stage at which unequal evaluation may emerge but does not, by itself, establish an unconscious psychological mechanism. Combining randomized decision environments, repeated choices, and behavioral measures of political learning connects appearance-based judgment to the acquisition and use of substantive information in the Colombian electoral context.

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